
Optimized Modeling and Predictive Analysis of Complex Systems

Summary

Problem Restatement. Clarify the **core problem** and **constraints**; define the **decision variables**, **objective function**, and **feasible region**.

Uniqueness of the Modeling Approach. Briefly describe the **model category** (e.g., differential equations, optimization, stochastic simulation) and the **unique assumptions** or **innovations** relative to existing methods.

Core Algorithm Logic. Outline the **solution procedure**: input, main iterative or recursive process, and output metrics; specify the **software/algorithm** used (e.g., Gurobi, Monte Carlo) if applicable.

Most Influential Quantitative Conclusions. Provide specific **numerical results** and **policy implications** (e.g., "cost reduction of 15.4%" or "service level exceeding 98%"); avoid vague wording. Optionally present findings from **sensitivity** or **robustness** analyses.

The above four modules use bolded terminology and short paragraphs to facilitate modular quick scanning, with clear paragraph spacing.

Keywords: Mathematical Modeling; Sensitivity Analysis; AI Tool Usage Statement.

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1 Introduction

1.1 Research Background and Motivation

Generative Artificial Intelligence (Gen-AI) is evolving from a niche auxiliary tool into an indispensable foundational resource across various industries. With its exponential growth in capabilities, Gen-AI is not only optimizing workflows but fundamentally reshaping the task structure of the labor market. Confronted with this radical technological transformation, higher education institutions (including comprehensive universities, vocational and technical colleges, and art academies) face unprecedented challenges: how to adjust talent cultivation models to ensure that their graduates possess enduring competitiveness and adaptability in the future workplace surrounded by AI?

AI Usage Statement. We disclose the usage of AI tools in this paper and in the **AI Tool Usage Report** at the end of the document. Translate the following LaTeX content:

1.2 Literature Review

Cite existing research to enhance the credibility of the model. [4]

1.3 Problem Restatement

Part I (Professional Level: "Future Evolution" under the Influence of Gen-AI) The problem requires us to select one specific profession from each of the three categories: **STEM, Trade/Vocational Education, and Arts**. Based on traceable data and cited sources, we are to characterize the future changes of these professions in the context of "Gen-AI proliferation." Here, "future" should be understood as the evolution of professional forms within a defined prediction window (e.g., the next several years), including the direction and magnitude of potential reductions, stability, or expansions in job content and demand. "Current trajectory" can be interpreted as the trend along which Gen-AI penetration and usage in the industry are advancing. "Expected impact" refers to the measurable changes in job structure and practitioners' work methods brought about by this penetration. All judgments must specify the data sources used and explain the key driving factors we believe will propel professional changes.

Part II (Educational Program Level: Providing Actionable Recommendations for Three Specific Institutional Programs) After completing the analysis at the occupational level, the problem further requires: selecting one "corresponding specific institution and specific program" for each occupation (from a university, a technical school/apprenticeship system, and an art school, respectively), and grounding the recommendations in the management decisions of these three programs. The recommendations should cover at least two core issues: first, whether the program scale should expand or contract in response to occupational changes (and how to attract more students during expansion or how to absorb potential students into other programs within the institution during contraction); second, what the program should "teach and how to handle the use of Gen-AI," such as different strategies like prohibition, limited allowance, or integration into the curriculum and training system. Here, "best support/best preparation" should be understood as: providing program decision-makers with **actionable and comparable** solutions, and using the output of mathematical models to furnish an evidence chain for these solutions, rather than remaining at the level of general principles.

Part III (Evaluation Scope: Beyond "Employability" and Proposed Extrapolation Boundaries) The problem statement notes that "graduate employability" may not be the sole criterion for evaluating policy success; thus, teams are required to propose and

justify additional evaluation dimensions, and explain how conclusions and recommendations would change when these dimensions are incorporated. To avoid conceptual ambiguity, "employability" can be understood as graduates' ability to obtain and maintain positions in a job market where Gen-AI is widely prevalent; while "other factors" may include, but are not limited to: the resource consumption costs (energy/water usage) of teaching and using Gen-AI, as well as attribution and integrity risks in academic and creative contexts. Finally, if a team believes that recommendations for a particular institution/program can be generalized to other institutions or programs, they must explicitly state the conditions and boundaries for such extrapolation, clarifying under which circumstances the generalization holds and under which it does not.

2 Assumptions and Notation

2.1 Basic Assumptions

Translate the following LaTeX content:

2.2 Symbol Definitions

All symbols used in this paper are defined in the following table.

Table 1: Symbol Definitions Used in This Paper

Symbol	Description	Domain
Basic Indices		
i	Occupation index	
j	Task index	
t	Time	$\mathbb{Y}$
N	Number of available tasks	
Occupation Choice and Feature Space		
x	Candidate occupation	
$f(x)$	Occupation feature vector ($digit(x)$, $physical(x)$, $iprisk(x)$)	$[0, 1]$
$digit(x)$	Digitalization level	$[0, 1]$
$physical(x)$	Physical anchoring level	$[0, 1]$
$iprisk(x)$	Intellectual property/attribution risk	$[0, 1]$
$D(x)$	Data availability score	$[0, 1]$
$\Delta(x_i, x_j)$	Mechanism comparison distance: $\ f(x_i) - f(x_j)\ _2$	
λ	Trade-off coefficient in occupation choice	Dimensionless
$v_m(x)$	Raw score of occupation x under O*NET descriptor m	$[0, 100]$
$z_m(x)$	Min-max normalized score	$[0, 1]$
Task Weights and Impact Parameters		
w_{ij}	Weight of task j in occupation i ; $\sum_j w_{ij} = 1$	$[0, 1]$
$w_{ij}(0)$	Baseline task weight (initial time)	$[0, 1]$
$w_{ij}(t)$	Time-varying task labor share (task structure)	$[0, 1]$

Continued on

Table 1 – Continued

Symbol	Description	Domain
s_{ij}	Substitution score for task j (degree to which GenAI reduces human effort)	$\{0.25, 0.5, 0.75, 1\}$
c_{ij}	Enhancement score for task j (degree to which GenAI expands functional demand)	$\{0.25, 0.5, 0.75, 1\}$
α_i	Occupation-level substitution effect: $\sum_j w_{ij}s_{ij}$	$[0, 1]$
β_i	Occupation-level enhancement effect: $\sum_j w_{ij}c_{ij}$	$[0, 1]$
μ_i	Net occupational impact: $\beta_i - \alpha_i$	$[-1, 1]$
$\bar{c}_i$	Occupational enhancement potential: $\sum_j w_{ij}(0)c_{ij}$	$[0, 1]$
Occupational Evolution Dynamics		
$E_i(t)$	Employment demand for occupation i at time t	Number of workers
$A(t)$	GenAI adoption trajectory (scenario family: fast/medium/slow)	$[0, 1]$
g_i	Non-AI baseline growth rate for occupation i	Year ⁻¹
k	Steepness parameter of the adoption curve	> 0
t_0	Inflection point time of the adoption curve (50% adoption)	Year
Task-Level Functional Demand Model		
$Q_i(t)$	Overall functional output demand index for occupation i	Dimensionless
$\ell_{ij}(t)$	Labor intensity of task j (human effort per unit output)	Dimensionless
$D_{ij}(t)$	Human labor demand index for task j	Dimensionless
κ_s	Substitution speed coefficient	> 0
κ_c	Enhancement factor amplifying demand	> 1
Occupational Redefinition Triggers		
$M_{\text{drop}}(t)$	Extinction mass (sum of baseline weights of tasks whose demand becomes negligible)	$[0, 1]$
ε	"Negligible task" threshold (e.g., 0.05 for 5% of baseline)	$(0, 1]$
θ	Extinction mass threshold triggering occupational redefinition	$(0, 1]$
δ	Task share drift threshold (Jensen–Shannon divergence)	> 0
t^*	Occupational redefinition trigger time	Year
$\text{JS}(\cdot\ \cdot)$	Jensen–Shannon divergence	$[0, 0.5]$
Mechanism Strength Diagnostics		
$\text{Var}_w(s)$	Weighted variance of substitution scores: $\sum_j w_j(s_j - \alpha)^2$	$[0, 1]$
$\text{Var}_w(c)$	Weighted variance of enhancement scores: $\sum_j w_j(c_j - \beta)^2$	$[0, 1]$
$h_s(g)$	Weight mass for substitution score tier g	$[0, 1]$
$h_c(g)$	Weight mass for enhancement score tier g	$[0, 1]$
$m(d)$	Weight mass for capability dimension d	$[0, 1]$
$\varepsilon_s, \varepsilon_c$	Minimum variance threshold (default 0.005)	$[0, 1]$
η_s, η_d	Maximum concentration threshold (default 0.70, 0.60)	$(0, 1]$
Authority Relabeling Process		
dim_j	Set of capability dimensions mapped to task j	Set of dimensions
d	Capability dimension index (e.g., code generation, debugging, testing)	Index
S_d	Substitution tier for dimension d	$\{0.25, 0.5, 0.75, 1\}$

Continued on

Table 1 – Continued

Symbol	Description	Domain
C_d	Enhancement tier for dimension d	$\{0.25, 0.5, 0.75, 1\}$
$\text{sim}(j, d)$	Semantic similarity between task j and dimension d	$[0, 1]$
τ	Semantic matching threshold	$[0, 1]$
π_{jd}	Assignment weight from task j to dimension d	$[0, 1]$
Task DNA Fields (O*NET Data)		
Task_j	Task statement text	Task statement
IM_j	Task importance rating	$[0, 1]$
FR_j	Task frequency rating	$[0, 1]$
$\tilde{w}_j$	Unnormalized weight: $\text{IM}_j \cdot \text{FR}_j$	$[0, 1]$
SOC	O*NET-SOC occupation code	Occupation code

3 Data Sources

3.1 Data Description and Task DNA Source

We extract task statements and task ratings for SOC 15-1252.00 from the O*NET 30.1 text database and calculate task weights based on Importance (IM) and Frequency of Task (FT) ratings. [7]

Table 2: Data Sources and Variables for Task DNA

Variable/Data	Description	Source
O*NET Task Statements	Task text and identifiers for SOC 15-1252.00	O*NET Database [7] 30.1
O*NET Task Ratings (IM, FT)	Importance and Frequency ratings used to calculate task weights	O*NET Database [7] 30.1
Task DNA CSV	Processed task table containing IM, FR, weights, and rule-based s/c	Derived from O*NET 30.1 [7]

4 Model Construction and Implementation

Translate the following LaTeX content:

4.1 Career Choice as a Modeling Decision

Translate the following LaTeX content: The premise of the problem requires selecting one profession each from the STEM, skilled trades, and arts fields. We do not treat this step as an ad hoc assumption (subjective choice), but explicitly model the profession selection as an optimization problem.

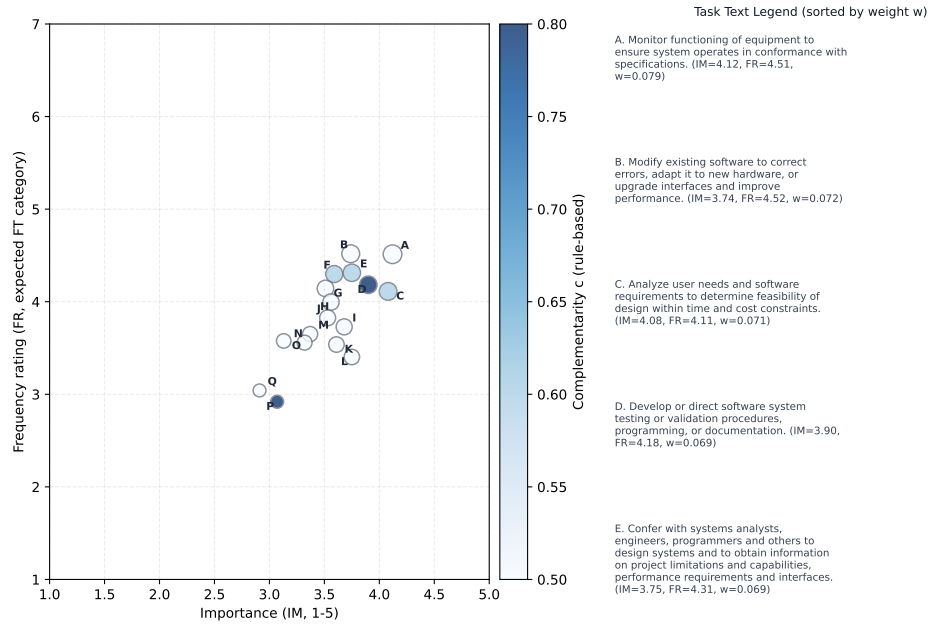


Figure 1: Task-level IM–FR exposure for software developers (SOC 15-1252.00). Bubble size encodes the normalized task weight w , color encodes rule-based complementarity c , and edge thickness reflects rule-based substitutability s . Letter labels map to task text in the legend panel.

This ensures that the subsequent analysis of AI-driven impacts and educational recommendations is built upon a reproducible and data-driven framework. The core objective of profession selection is not representativeness of all professions, but *explanatory contrast*. Specifically, the task structures of the three professions should exhibit significant differences in how they interact with GenAI. In addition to mechanistic contrast, the problem explicitly emphasizes data-driven reasoning and subsequent policy recommendations. Therefore, the selected professions must also have comprehensive data support and policy relevance.

4.1.1 Objective Parameterization of Occupational Characteristics Based on O*NET

Translate the following LaTeX content: To avoid subjective assignment of the occupational characteristic parameters $digit(x)$, $physical(x)$, and $iprisk(x)$, we adopt the **O*NET** occupational database maintained by the U.S. Department of Labor as a unified data source. Detailed information on the data sources is summarized in Section 3. Based on expert surveys and practitioner feedback, O*NET provides a large number of *numerical scale descriptors* for each occupation. Most indicators are naturally given as real numbers between 0 and 100, making them suitable for direct use in quantitative modeling. We do not attempt to find a "single correct" metric for each dimension. For each feature axis, we select a small number of semantically directly corresponding and downloadable O*NET descriptors and use them as proxy variables for that axis. Based on the problem statement and data availability constraints, we identify three dominant dimensions that govern the heterogeneity of AI impacts across occupations:

- **Digitalization degree** $digit(x)$: The extent to which core tasks can be represented, automated, or enhanced digitally;

- **Physical anchoring** $physical(x)$: The degree of reliance on on-site physical presence or manual interaction with the environment for task execution;
- **Intellectual property and attribution risk** $iprisk(x)$: The sensitivity of value creation to authorship, originality, and ownership.

Each candidate occupation x is represented by a normalized three-dimensional feature vector

$$f(x) = (digit(x), physical(x), iprisk(x)) \in [0, 1]^3, \quad (1)$$

where higher values indicate a stronger presence of the corresponding feature. For three occupational categories (x_1, x_2, x_3) sampled from STEM, skilled trades, and arts respectively, we define the contrast between two occupations as the Euclidean distance between their feature vectors:

$$\Delta(x_i, x_j) = \|f(x_i) - f(x_j)\|_2. \quad (2)$$

Define the data availability score $D(x_k)$, which reflects the ease and richness of obtaining data related to that occupation. The overall selection objective is formulated as

$$\max_{x_1 \in STEM, x_2 \in Trade, x_3 \in Arts} \left(\sum_{i < j} \Delta(x_i, x_j) + \lambda \sum_{k=1}^3 D(x_k) \right), \quad (3)$$

where λ is a weighting coefficient that balances interpretability comparison with empirical feasibility. This formula does not claim global optimality across all possible careers. Instead, it generates a carefully constructed set of high-contrast careers that maximizes analytical clarity under realistic data and time constraints.

Normalization and Reproducibility For each occupation x , let $v_m(x)$ denote its raw score (0–100) under the O*NET descriptor m . We apply min-max normalization across the entire set of occupations:

$$z_m(x) = \frac{v_m(x) - \min_y v_m(y)}{\max_y v_m(y) - \min_y v_m(y)} \in [0, 1], \quad (4)$$

and take the equally weighted average of multiple descriptors under the same feature axis to obtain $digit(x)$, $physical(x)$, and $iprisk(x)$. This process is entirely based on publicly available data and explicit rules, ensuring the reproducibility and auditability of the parameter values. We visualized the three-dimensional heterogeneity and its two-dimensional projections to verify that the selected occupations occupy distinct regions in the feature space (Figure 2 and Figure 3).

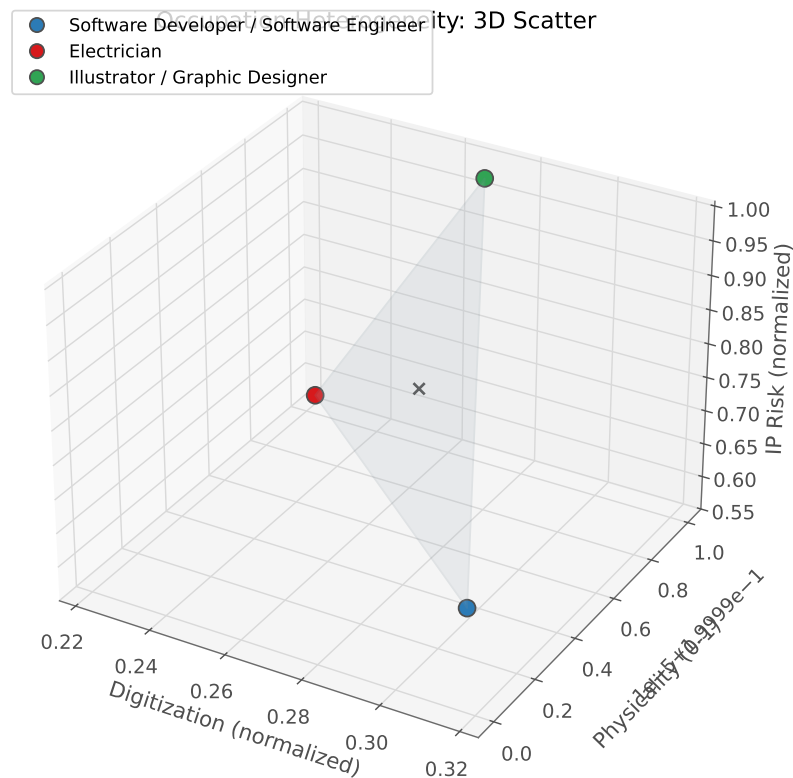


Figure 2: Three-dimensional heterogeneity of selected STEM, skilled trade, and artistic occupations based on O*NET descriptors (globally normalized).

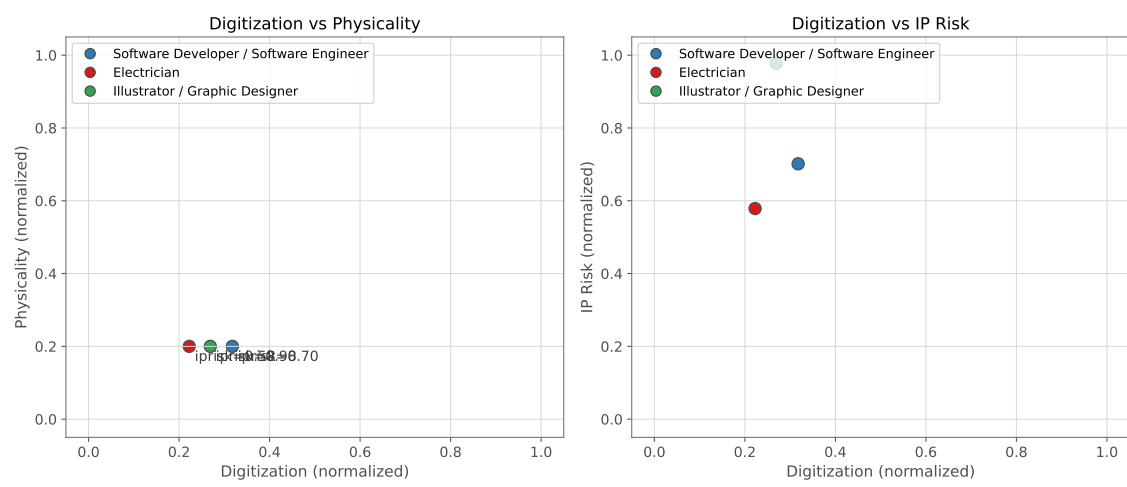


Figure 3: Two-dimensional projection of the heterogeneous space; annotations report the values of the remaining axes to enhance interpretability (global normalization).

4.1.2 Revision and Final Configuration of Selected Careers

Translate the following LaTeX content: In the initial scheme, we selected structural/civil engineers as the STEM profession. However, upon further examination of the distribution of the three professions along the dimensions of "task structure" and "physical constraints," we found that this choice exhibits some similarity with the trade profession (electricians) on the "physical anchoring strength" axis, which may weaken the contrast among the three professions in the mechanism space. To maximize the heterogeneity of the task structure and enhance the identifiability of subsequent GenAI impact mechanisms, we revise the STEM endpoint by replacing it with **software developers/software engineers**. This revision does not alter the structure of the original selection model; it only adjusts the positions of occupational endpoints in the feature space: software developers exhibit an extremely high degree of digitalization and an extremely low level of physical anchoring, thereby significantly increasing their distance from electricians in the feature vector $f(x)$ and enhancing the overall contrastive objective function value. It must be emphasized that we do not assume that the software development profession will be simply "replaced" by GenAI. Instead, this profession is more likely to undergo a **restructuring of task composition**: the proportion of coding and boilerplate implementation decreases, while the proportion of tasks such as requirement clarification, system architecture design, verification testing, and security and compliance responsibilities increases. It is precisely this shift from "execution-oriented tasks" to "verification and responsibility-oriented tasks" that makes software development an ideal STEM endpoint for studying the impact mechanisms of GenAI. After completing this revision, the final configurations of the three selected occupations are as follows:

- **STEM: Software Developer/Software Engineer.** High degree of digitalization, low physical anchoring; GenAI can directly intervene in core tasks, but primarily manifests as task redistribution rather than occupational obsolescence.
- **Skilled Trade: Electrician.** High physical anchoring, strong on-site constraints; AI mainly serves as a planning, diagnostic, and assistive tool, making it difficult to replace core operational tasks.
- **Arts: Commercial Illustrator/Graphic Designer.** High sensitivity regarding intellectual property and attribution; GenAI directly influences the form of output and the mechanisms of value pricing.

4.2 Model Selection for Career Evolution

Translate the following LaTeX content: The second core requirement of the problem is to construct a *data-driven* model for the future evolution of professions. Before introducing formal equations, we first define the meaning of "future evolution" in a computable sense, and then systematically evaluate alternative modeling approaches, ensuring that the model selection itself does not become an implicit assumption.

4.2.1 Computable Definition of Future Evolution

Translate the following LaTeX content: We characterize the future evolution of occupation i through two primary outputs:

- **Occupational Demand:** The temporal evolution of employment demand $E_i(t)$ over a 10-year horizon under rapid, moderate, and slow GenAI adoption scenarios;
- **Task Structure:** The reallocation of task weights $T_i(t)$ within the occupation, indicating which tasks expand, contract, or are restructured.

This definition ensures that professional-level outcomes stem from task-level mechanisms, rather than being directly imposed.

4.2.2 Comparison of Candidate Modeling Methods

Translate the following LaTeX content: We consider three representative approaches for modeling career evolution:

Method A: Task Aggregation Impact Model This method treats occupations as weighted combinations of tasks. The substitution and enhancement effects driven by GenAI are specified at the task level and then aggregated to the occupational level using task weights. Most parameters can be constructed from O*NET task data [7] and publicly available AI exposure scales [4], ensuring transparency and reproducibility.

Method B: Time Series Model for Recruitment Information This method infers occupational changes from trends in AI-related keywords and skill requirements in job postings. While closely tied to market signals, it faces identification challenges and high data processing overhead, making it unsuitable as the primary modeling framework.

Method C: Agent-Based Simulation Agent-based models simulate strategic interactions among firms, workers, and educational institutions. Although rich in narrative, they require extensive calibration and introduce substantial model complexity, which conflicts with transparency requirements under strict page limits.

4.2.3 Final Model Selection

Translate the following LaTeX content: Based on the above comparison, we adopt the Task Aggregation Impact Model (Method A) as the primary framework. The motivation for this selection is:

- **Mechanism Clarity:** Occupational changes are explicitly derived from task-level effects;
- **Data Alignment:** Key parameters are based on publicly available occupational data [7, 9];
- **Controllable Complexity:** The model maintains interpretability, making it suitable for scenario and sensitivity analysis;
- **Transferability:** The framework can be easily extended to other occupations or educational pathways [6].

Method B is reserved for potential validation or supplementary analysis, while Method C is excluded from the primary analysis due to calibration and transparency issues.

4.3 Dynamic Model of Task Aggregation for Occupational Evolution

Translate the following LaTeX content: After justifying the task aggregation approach, we now present its mathematical formulation. We adopt a task-to-occupation framework with minimal structure: GenAI adoption $A(t)$ enters as an exogenous driver, substitution and enhancement are specified at the task level, and the net occupational impact is aggregated via task weights, yielding the dynamic trajectory of occupational demand $E_i(t)$.

4.3.1 Task Aggregation and Occupational Dynamics

Translate the following LaTeX content: Let occupation i consist of tasks $j = 1, \dots, J_i$ with weights w_{ij} satisfying

$$\sum_{j=1}^{J_i} w_{ij} = 1. \quad (5)$$

For each task j , let $s_{ij} \in [0, 1]$ denote the substitution intensity and $c_{ij} \in [0, 1]$ denote the enhancement (complementarity) intensity. Define the occupation-level aggregate effects as

$$\alpha_i = \sum_{j=1}^{J_i} w_{ij} s_{ij}, \quad \beta_i = \sum_{j=1}^{J_i} w_{ij} c_{ij}. \quad (6)$$

Let $E_i(t)$ be the employment demand for occupation i at time t , and g_i capture the baseline growth/decay from non-AI factors (estimated from public trends or set to 0 under scenario analysis). Given a family of adoption trajectory scenarios $A(t) \in [0, 1]$ (fast/medium/slow), we model occupational evolution as

$$\frac{dE_i(t)}{dt} = E_i(t) \left(g_i + A(t) (\beta_i - \alpha_i) \right). \quad (7)$$

This formulation ensures that occupational outcomes stem from task-level mechanisms, supporting interpretability, reproducibility, and data-driven reasoning.

4.4 GenAI Capability Dimension Scoring from Authoritative Reports: Scale Definition and Evidence Narrative

Translate the following LaTeX content: To replace the subjective "keyword tagging" approach, we adopt capability scores derived from authoritative evaluations and official technical reports. These scores quantify GenAI's **substitution** and **enhancement** capabilities across key dimensions of software development tasks, providing an auditable basis for mapping software developer task DNA to (s, c) inputs.

4.4.1 Scale Definition

Translate the following LaTeX content: We use three levels (low/medium/high) to represent GenAI capabilities, encoded as 0.25, 0.50, and 0.75:

- **Low (0.25):** GenAI has limited autonomous capability in this task type and cannot reliably deliver satisfactory results without human intervention; it can only replace a small portion of the work but can still provide assistance under human supervision.

- **Medium (0.50):** GenAI performs moderately, completing some tasks or providing useful suggestions, but still requires human review and collaboration; it cannot independently replace human execution.
- **High (0.75):** GenAI performs strongly, approaching professional human levels in many subtasks; with appropriate constraints and validation, it can handle most of the workflow and provide significant enhancement, though it is still not perfect.

Note: The values of 0.25/0.50/0.75 are rough modeling encodings rather than precise measurements. All levels are based on publicly released reports and comparative experiments from the past two years.

4.4.2 Evidence of Capabilities by Task Type (Narrative Argument)

Translate the following LaTeX content:

Code Generation **Substitution: High (0.75); Enhancement: High (0.75).** GenAI has demonstrated strong performance in automated code generation. OpenAI GPT-4 achieves 67.0% pass@1 on HumanEval, significantly higher than GPT-3.5's 48.1%; Anthropic Claude 2 reaches 71.2% on the same benchmark. This indicates that leading models can correctly solve over half of typical coding problems. Google CEO Sundar Pichai stated that approximately 25% of Google's new code is generated by AI tools, reflecting a significant productivity boost: given an intent or comment, models can generate a large amount of draft code and accelerate development. For simple modules, models can partially substitute manual coding. However, outputs still require human review and testing to reduce logical errors and security issues. Overall, coding/implementation is one of the most substitutable and enhancing domains for GenAI.

Testing and Verification **Substitution: Medium (0.50); Enhancement: High (0.75).** GenAI performs moderately in generating test cases and supporting verification, but is improving. Research reports indicate that LLM-generated unit tests can achieve coverage comparable to advanced automated tools and are often more readable. For example, experiments guided by static analysis at IBM Research found that GPT-based systems can generate competitive Java/Python unit tests in terms of coverage and defect detection capability, while producing developer-friendly tests. This supports strong enhancement in routine function-level verification. However, generated tests may target irrelevant scenarios or have weak assertions, requiring manual review; thus, full substitution remains difficult. In practice, GenAI can significantly reduce manual test writing and accelerate result checking, while humans verify test correctness.

Debugging and Error Fixing **Substitution: Low (0.25); Enhancement: Medium (0.50).** End-to-end debugging remains a weakness: autonomous success rates are relatively low, but the auxiliary value is evident. A 2025 Microsoft Research study reported that even with debugger tool access, the best model (Anthropic Claude 3.7 Sonnet) solved only 48.4% of 300 curated debugging tasks on average, while other models achieved around 30%. This indicates that GenAI cannot fully replace manual debugging. Limitations include ineffective use of debugging signals, poor strategy selection, and a lack of step-by-step reasoning traces. Therefore, substitution is rated low. Meanwhile, enhancement is evident: with logs or exception traces, models can propose

useful fixes; in some setups, "self-debugging" via few-shot demonstrations improves success rates by approximately 2%–12%. In practice, GenAI serves more as an assistant than an independent "fixer."

System Design/Architecture **Substitution: Low (0.25); Enhancement: Medium (0.50).** Architecture and system design require global reasoning and responsible trade-offs; GenAI currently provides limited support. While there is no single authoritative benchmark that directly measures system design capability, models typically produce plausible but incomplete plans, omitting critical constraints or relying on incorrect assumptions. Since system design has no single correct answer and requires responsible decision-making, autonomous substitution is rated low. Nevertheless, enhancement can be meaningful: models can draft design documents, summarize patterns, and brainstorm alternatives as a starting point for architects, improving the efficiency of discussions.

Demand Analysis **Substitution: Low (0.25); Enhancement: Medium (0.50).** Requirement work requires deep domain background and ambiguity resolution. GenAI can assist in summarizing and structuring text but cannot replace analysts. Although specialized evaluations are limited, related tasks such as natural language to formal specifications provide evidence: on Spider (natural language to SQL), even with enhancement techniques, GPT-4's improvement is only about 2%–3%, and complex queries remain challenging. This indicates that understanding complex, ambiguous requirements is unreliable, hence substitution is low. However, enhancement is moderate: models can summarize lengthy documents, generate user story templates, ask clarifying questions, and accelerate requirement organization under human supervision.

User Communication and Documentation **Substitution: Medium (0.50); Enhancement: High (0.75).** GenAI excels in language generation, strongly enhancing user communication and technical documentation. Models such as GPT-4 and Claude 2 can generate coherent long-form drafts and adhere to structured formats (e.g., Markdown/JSON), enabling rapid drafting of API documentation, manuals, and user responses. Many routine user queries can be answered fluently and consistently. However, risks of inaccuracy and hallucination persist, especially for system-specific details, making full substitution of human communication still risky; substitution is rated as medium while enhancement is rated as high.

4.5 Task Impact-Driven Relabeling via Authoritative Assessment: From Keyword Tags to Competency Dimension Allocation

Translate the following LaTeX content: To enhance the defensibility and reproducibility of the task impact parameters (s_{ij}, c_{ij}) , we abandon subjective keyword tagging and adopt a relabeling workflow of "**Authoritative Assessment** $\rightarrow$ **Competency Dimensions** $\rightarrow$ **Semantic Matching** $\rightarrow$ **Segment Assignment**". The core idea is to extract a set of *competency dimensions* (such as code generation, debugging, testing/verification, requirements analysis, system design, documentation/communication) with alternative/enhanced levels from official evaluation reports, map each software developer task to 1–2 competency dimensions, and then assign (s_j, c_j) on a discrete grid.

Capability Mapping and Traceability Requirements For each task j , define its mapped ability set dim_j (typically 1–2 dimensions to maintain interpretability). Task-level substitution and enhance-

Table 3: GenAI Capability Rubric from Authoritative Evaluation Reports (Used for Mapping Software Developer Tasks to *s/c*)

Task Type	Substitution Score	Augmentation Score	Key Evidence and Support
Code Generation	High (0.75)	High (0.75)	GPT-4 achieves 67.0% pass@1 on HumanEval [8]. Claude 2 achieves 71.2% [2]. Google reports approximately 25% of new code is AI-generated [10].
Testing and Verification	Medium (0.50)	High (0.75)	LLM-generated unit tests can achieve competitive coverage and readability. Research shows higher test quality with static analysis guidance [1]. Human review is still required, hence substitution remains medium.
Debugging and Bug Fixing	Low (0.25)	Medium (0.50)	Microsoft Research reports end-to-end debugging success rates below 50% even with debugging tools (best model 48.4%) [10]. Prompted fix suggestions are helpful; "self-debugging" can improve accuracy by up to 12 percentage points [3].
System Design	Low (0.25)	Medium (0.50)	Complex architectures require comprehensive reasoning; current GenAI often produces partial solutions [8]. Direct quantitative benchmarks are limited, but models can assist in drafting and documentation to accelerate design discussions [5].
Requirements Analysis	Low (0.25)	Medium (0.50)	Models show limited improvement on NL→spec tasks (e.g., 2–3% on Spider) [3]. However, summarization and question generation can improve requirements elicitation and analysis efficiency [8].
User Communication and Documentation	Medium (0.50)	High (0.75)	Models like Claude 2 can efficiently generate long-form structured documentation and answer common questions [2]. Accuracy remains an issue; critical content still requires human verification [8].

ment are assigned on a discrete scale:

$$s_j, c_j \in \{0.25, 0.5, 0.75\}. \quad (8)$$

To meet the "data-driven" audit criterion, each task must carry a **traceability chain**: (i) the mapped ability dimension dim_j , (ii) at least one authoritative source link, (iii) a brief explanation of the task-to-dimension matching logic. This ensures that (s_j, c_j) are reviewable rather than based on researcher intuition.

Semantic Thresholds, Anomaly Reporting, and Limited Human Coverage Since semantic matching may lead to false negatives and false positives, the matching threshold is considered a controllable decision parameter: A threshold that is too high may miss valid matches; one that is too low may introduce erroneous matches. Therefore, the relabeling output must include similarity scores and anomaly reports (such as low-similarity matches, multi-dimensional assignment conflicts), and allow for small-scale manual override lists for a few semantically ambiguous high-weight tasks. This maintains overall reproducibility while preventing a small number of mislabeled entries from skewing the professional-level aggregated values (α_i, β_i) .

4.6 Aggregating Task DNA into Career-Level Impact Parameters: Software Developers

Translate the following LaTeX content: This section aggregates the task-level inputs (w_{ij}, s_{ij}, c_{ij}) of software developer task DNA into career-level impact parameters $(\alpha_i, \beta_i, \mu_i)$, which serve as inputs to the evolution equation $\frac{dE_i(t)}{dt} = E_i(t)(g_i + A(t)(\beta_i - \alpha_i))$.

4.6.1 Top- N Availability and Aggregation

Translate the following LaTeX content: In practice, O*NET task ratings may be missing for certain tasks. Therefore, we adopt a "Top- N_i available tasks" strategy: for occupation i , we only retain tasks with sufficient ratings to construct weights, and renormalize so that $\sum_{j=1}^{N_i} w_{ij} = 1$. The occupation-level substitution pressure, augmentation benefit, and net direction are defined as

$$\alpha_i = \sum_{j=1}^{N_i} w_{ij} s_{ij}, \quad \beta_i = \sum_{j=1}^{N_i} w_{ij} c_{ij}, \quad \mu_i = \beta_i - \alpha_i. \quad (9)$$

4.6.2 Interpretability Decomposition: Top- K Contribution Lower Bound

Translate the following LaTeX content: To enhance interpretability, we decompose the contributions of high-weight tasks. Define partial sums

$$\alpha_i^{(K)} = \sum_{j \in \text{Top-}K} w_{ij} s_{ij}, \quad \beta_i^{(K)} = \sum_{j \in \text{Top-}K} w_{ij} c_{ij}. \quad (10)$$

For the currently available Top-5 samples, $\sum_{j \in \text{Top-}5} w_{ij} = 0.360897$, the partial contributions $\alpha_i^{(5)} = 0.1804485$ and $\beta_i^{(5)} = 0.2153186$, indicating that high-weight tasks (typically testing/maintenance/requirements/collaboration) will dominate the aggregated impact.

4.6.3 Validation of Scoring Rules: Avoiding Near-Constant s/c

Translate the following LaTeX content: If many tasks share the same discrete scores (such as 0.50), then α_i and β_i become weakly discriminative, weakening the mechanism's interpretability. Therefore, we examine the concentration of weight quality on the discrete grid $\mathcal{G} = \{0, 0.25, 0.5, 0.75, 1\}$:

$$p_s(g) = \sum_{j: s_{ij}=g} w_{ij}, \quad p_c(g) = \sum_{j: c_{ij}=g} w_{ij}, \quad g \in \mathcal{G}, \quad (11)$$

using concentration and normalized entropy:

$$C_s = \max_{g \in \mathcal{G}} p_s(g), \quad C_c = \max_{g \in \mathcal{G}} p_c(g), \quad (12)$$

$$H_s = -\frac{\sum_{g \in \mathcal{G}} p_s(g) \ln p_s(g)}{\ln |\mathcal{G}|}, \quad H_c = -\frac{\sum_{g \in \mathcal{G}} p_c(g) \ln p_c(g)}{\ln |\mathcal{G}|}. \quad (13)$$

High C_s or low H_s indicates score compression and triggers minimal calibration.

4.6.4 Minimum Calibration: Category Defaults with Limited Human Review

Translate the following LaTeX content: To avoid recollecting data, we calibrate s/c using only a small set of task categories, while keeping the task list and weights w_{ij} unchanged. Let $\kappa(j)$ map each task to a coarse category (e.g., requirement/stakeholder, coding/implementation, testing/verification/security, monitoring/operations) with default (s, c) pairs. We then manually review only a small number of high-weight tasks and record brief evidence descriptions, improving the identifiability of (α_i, β_i) while maintaining reproducibility and transparency.

5 Mechanism Power Test and Professional-Level Aggregation (Software Developers) after Authority Relabeling

Translate the following LaTeX content: After upgrading (s_j, c_j) from subjective keyword tags to the "authoritative assessment $\rightarrow$ capability dimension $\rightarrow$ semantic matching $\rightarrow$ segment allocation" process, we must verify the **mechanism strength**: the task-level heterogeneity of (s_j, c_j) should be sufficiently substantial, and the task weights w_j should not collapse (α, β) into a single default-driven outcome. Otherwise, even authoritative sources will yield weak identifiability in the evolutionary model.

5.1 Professional-Level Impact Aggregation Value

Translate the following LaTeX content: For the available task set (with size N , allowing $N < 20$), after renormalization such that $\sum_{j=1}^N w_j = 1$, we define

$$\alpha = \sum_{j=1}^N w_j s_j, \quad \beta = \sum_{j=1}^N w_j c_j, \quad \mu = \beta - \alpha. \quad (14)$$

5.2 Diagnosis: Heterogeneity and Effective Diversity

Translate the following LaTeX content:

(1) Weighted Variance (Heterogeneity)

$$\text{Var}_w(s) = \sum_{j=1}^N w_j (s_j - \alpha)^2, \quad \text{Var}_w(c) = \sum_{j=1}^N w_j (c_j - \beta)^2. \quad (15)$$

Low variance indicates that the score compression or mapping rules are overly convergent, weakening identifiability.

(2) Weighted Hierarchical Histogram (Avoiding Single-Level Dominance) Let $\mathcal{G}_{LMH} = \{0.25, 0.5, 0.75\}$. Define the hierarchical weight mass

$$h_s(g) = \sum_{j: s_j=g} w_j, \quad h_c(g) = \sum_{j: c_j=g} w_j, \quad g \in \mathcal{G}_{LMH}. \quad (16)$$

A larger $\max_g h_s(g)$ or $\max_g h_c(g)$ indicates a default-driven collapse.

(3) Quality of Capability Dimension Weights (Avoiding Weight Concentration) For each task j , let dim_j be the set of dimensions to which it is mapped (1–2 dimensions). If multiple mappings exist, let π_{jd} assign task j to dimension d , satisfying $\sum_{d \in \text{dim}_j} \pi_{jd} = 1$. The dimensional weight mass is

$$m(d) = \sum_{j=1}^N w_j \pi_{jd}. \quad (17)$$

A high $\max_d m(d)$ indicates low effective diversity despite broad nominal coverage.

5.3 Engineering Threshold and Action (Gating before $E(t)$)

Translate the following LaTeX content: We use engineering thresholds as gates to determine whether to proceed with scenario evolution:

$$\text{Var}_w(s) \geq \varepsilon_s, \quad \text{Var}_w(c) \geq \varepsilon_c, \quad \max_g h_s(g) \leq \eta_s, \quad \max_d m(d) \leq \eta_d, \quad (18)$$

with default values $\varepsilon_s = \varepsilon_c = 0.005$, $\eta_s = 0.70$, $\eta_d = 0.60$. Violation indicates high mechanism risk, requiring optimization of the hierarchical partitioning or the mapping from dimensions to (s, c) before advancing to the scenario-driven $E(t)$ dynamics (without changing w_j).

6 Consistency Check and Instantiation of Evolution Equations: Software Developers' (α, β, μ)

Translate the following LaTeX content: After relabeling the task impacts through the process of "authoritative assessment $\rightarrow$ capability dimension $\rightarrow$ semantic matching $\rightarrow$ discrete level (s, c) ", we aggregate the task DNA back into the occupational-level impact parameters (α, β, μ) and first conduct a consistency check, to prevent downstream dynamics from depending on incorrectly implemented net impact terms.

6.1 Aggregate Values and Consistency Test

Translate the following LaTeX content:By definition,

$$\mu = \beta - \alpha. \quad (19)$$

Given the preview values $\alpha = 0.379113$ and $\beta = 0.629112$, we must have

$$\mu = 0.629112 - 0.379113 = 0.249999. \quad (20)$$

However, the preview JSON reports $\mu = 0.504112$, which contradicts the model definition. Therefore, before using μ as input to $E(t)$, the aggregation output must explicitly declare `mu_definition` and correct the implementation; otherwise, the evolutionary inference will systematically produce bias.

6.2 Mechanism Strength (Heterogeneity) Test: No Score Collapse

Translate the following LaTeX content:Using weighted variance

$$\text{Var}_w(s) = \sum_{j=1}^N w_j (s_j - \alpha)^2, \quad \text{Var}_w(c) = \sum_{j=1}^N w_j (c_j - \beta)^2, \quad (21)$$

the preview report shows $\text{Var}_w(s) = \text{Var}_w(c) = 0.015608$, exceeding the engineering threshold (e.g., 0.005). This indicates that the relabeled (s, c) retains sufficient heterogeneity; mechanism identifiability is not collapsed.

6.3 Minimum Sensitivity of Low-Confidence Tasks (Upper Bound Argument)

Translate the following LaTeX content:For low-confidence matches (e.g., `task_id=21666`, `weight` $w = 0.05434$), robustness can be argued via worst-case bound reasoning. Since the levels are $\{0.25, 0.5, 0.75\}$, the maximum swing per task level is 0.50 ($0.25 \leftrightarrow 0.75$), thus

$$\Delta_{\max}(\alpha) \leq w \cdot 0.50 = 0.05434 \times 0.50 = 0.02717, \quad (22)$$

similarly $\Delta_{\max}(\beta) \leq 0.02717$. Two lightweight scenarios (deleting the task; remapping to the suboptimal dimension) suffice to support a defensible sensitivity statement.

6.4 Evolution into the Profession after Adjusting μ

Translate the following LaTeX content:Once μ is corrected, the evolution equation becomes

$$\frac{dE(t)}{dt} = E(t)(g + A(t)\mu), \quad (23)$$

where $A(t)$ is the adoption scenario (fast/medium/slow), and g is the non-AI baseline (estimated from public trends or set to 0 under the relative index convention). The relative scale has a closed-form solution

$$\frac{E(t)}{E(t_0)} = \exp\left(\int_{t_0}^t (g + A(\tau)\mu) d\tau\right). \quad (24)$$

Therefore, before μ is consistently defined and $A(t)/g$ are explicitly specified, we do not proceed to the $E(t)$ scenario curves, avoiding uncomputable narratives.

7 Task-Level Functional Demand Model and Occupational Re-definition

7.1 Task-Level Functional Demand Model (Replacing Occupational-Level Job Demand)

7.1.1 Modeling Objectives and Notation

We do not predict occupational-level job counts, but rather predict how a profession's *task composition* evolves under GenAI adoption, and how the *human labor demand* associated with each task changes over time. The core proposition is that when a sufficiently large portion of a profession's baseline task bundle becomes negligible, the nature of that profession has changed and it should be redefined (the exact new job title is not central to the model).

Consider a profession i decomposed into tasks $j = 1, \dots, J_i$. For each task, we assume the following inputs have been constructed through the task DNA and authority relabeling process:

- Baseline task weight $w_{ij}(0)$, satisfying $\sum_j w_{ij}(0) = 1$.
- Substitution score $s_{ij} \in [0, 1]$: the extent to which GenAI can reduce human effort for task j .
- Augmentation score $c_{ij} \in [0, 1]$: the extent to which GenAI can expand the *functional* scope/demand associated with task j .
- Adoption curve $A(t) \in [0, 1]$: the proportion/intensity of GenAI adoption at time t (scenario-driven).

We introduce four state variables:

- $Q_i(t)$: Overall functional output demand index for profession i ,
- $\ell_{ij}(t)$: Labor intensity (human effort per unit output) for task j ,
- $w_{ij}(t)$: Time-varying task labor share (task structure),
- $D_{ij}(t)$: Human labor demand index for task j .

Intuitively, s_{ij} controls how quickly human effort becomes unnecessary for a task (automation channel), while c_{ij} controls how much overall functional demand can expand (scale-up channel). These are distinct mechanisms and are not forced to offset.

7.1.2 Dual-Channel Dynamics: Automation and Scale-Up

(a) Automation Channel: Substitution Reduces Labor Intensity. We assume that as GenAI adoption increases, tasks with higher substitution scores require less human effort per unit output:

$$\frac{d \ln \ell_{ij}(t)}{dt} = -\kappa_s A(t) s_{ij}, \quad \kappa_s > 0. \quad (25)$$

The closed-form solution is

$$\ell_{ij}(t) = \exp\left(-\kappa_s s_{ij} \int_0^t A(\tau) d\tau\right). \quad (26)$$

Parameter Meaning (Plain Language). κ_s is the *substitution speed coefficient*. A larger κ_s means human effort declines faster for the same $A(t)$ and s_{ij} .

(b) Task Structure Update: Intra-Occupational Reallocation. Automation changes the internal composition of a profession. We form unnormalized shares and renormalize:

$$\tilde{w}_{ij}(t) = w_{ij}(0) \ell_{ij}(t), \quad w_{ij}(t) = \frac{\tilde{w}_{ij}(t)}{\sum_k \tilde{w}_{ik}(t)}. \quad (27)$$

Interpretation. Highly automated tasks (small $\ell_{ij}(t)$) lose share; remaining human labor shifts to harder-to-substitute tasks.

(c) Scale-Up Channel (Optional): Augmentation Expands Overall Functional Demand. To capture the possibility that productivity tools may increase the total amount of work organizations choose to perform, we model the overall functional demand index:

$$\frac{d \ln Q_i(t)}{dt} = g_i + \kappa_c A(t) \bar{c}_i, \quad (28)$$

where

$$\bar{c}_i = \sum_j w_{ij}(0) c_{ij}. \quad (29)$$

Parameter Meaning (Plain Language). g_i is the baseline (non-GenAI) growth in functional demand; set $g_i = 0$ if one wishes to avoid macroeconomic assumptions. $\kappa_c \geq 0$ is the *amplification factor of augmentation on demand*. $\bar{c}_i$ summarizes the profession's augmentation potential under its baseline composition. (A variant uses $w_{ij}(t)$ instead of $w_{ij}(0)$; we use (29) for simplicity and parameter stability.)

(d) Task Labor Demand Index. We allocate functional demand to tasks via the current task structure:

$$D_{ij}(t) = Q_i(t) w_{ij}(t), \quad \frac{D_{ij}(t)}{D_{ij}(0)} = \frac{Q_i(t) w_{ij}(t)}{w_{ij}(0)}. \quad (30)$$

Interpretation. Even if overall demand $Q_i(t)$ grows, human demand for a task can still shrink if its task share $w_{ij}(t)$ collapses due to substitution.

7.1.3 Adoption Scenario $A(t)$

We use a parsimonious, interpretable logistic family:

$$A(t) = \frac{1}{1 + \exp(-k(t - t_0))}. \quad (31)$$

Parameter Meaning (Plain Language). k controls steepness (how fast adoption accelerates). t_0 is the inflection time when adoption reaches 50%. We define three scenarios (fast/medium/slow) by choosing three pairs (k, t_0) ; this keeps the number of drivers small while supporting sensitivity comparisons.

7.1.4 Occupational Redefinition Trigger (Change in Nature of Work)

We formalize "change in the nature of work" via *task extinction* and *task structure drift*. Either of two criteria can trigger redefinition.

(a) Extinction Mass (Task Demand Becomes Negligible). A task is considered functionally negligible when its human demand index falls below a small fraction of its baseline:

$$\frac{D_{ij}(t)}{D_{ij}(0)} < \varepsilon, \quad (32)$$

where $\varepsilon \in (0, 1)$ (e.g., $\varepsilon = 0.05$). Define the baseline-weighted extinction mass:

$$M_{\text{drop}}(t) = \sum_{j: D_{ij}(t)/D_{ij}(0) < \varepsilon} w_{ij}(0). \quad (33)$$

We consider a profession to have changed in essence if and only if

$$M_{\text{drop}}(t) \geq \theta, \quad (34)$$

where $\theta \in (0, 1)$ (e.g., $\theta = 0.30$) is the share of baseline task mass that must become negligible before triggering redefinition.

(b) Task Structure Drift (Distributional Change). Even without many tasks becoming fully negligible, a profession may become qualitatively different if its task share distribution changes sufficiently. Let $w_i(t)$ denote the vector $(w_{i1}(t), \dots, w_{iJ_i}(t))$. We use Jensen–Shannon divergence as a symmetric, bounded measure of drift:

$$\text{JS}(w_i(t) \parallel w_i(0)) \geq \delta, \quad (35)$$

where $\delta > 0$ is a conservative threshold chosen for robustness.

Trigger Time. Define

$$t^* = \inf \{t : M_{\text{drop}}(t) \geq \theta \text{ or } \text{JS}(w_i(t) \parallel w_i(0)) \geq \delta\}. \quad (36)$$

We report whether t^* exists within the forecast horizon and use it as model-based evidence that the profession should be redefined.

7.1.5 Computational Algorithm (Reproducible Process)

For each profession i and each adoption scenario:

1. Compute $A(t)$ from (31), and the cumulative adoption integral $I(t) = \int_0^t A(\tau) d\tau$ (numerical quadrature).
2. Compute labor intensities $\ell_{ij}(t)$ using (26).
3. Update task shares $w_{ij}(t)$ via (27).
4. Solve for $Q_i(t)$ using (28). If avoiding macroeconomic assumptions, set $g_i = 0$ and $\kappa_c = 0$, so that $Q_i(t) \equiv 1$.
5. Compute task labor demand indices $D_{ij}(t)$ from (30).
6. Evaluate redefinition criteria (34) and (35), obtaining t^* via (36) if it exists.

7.1.6 Parameter Table (To Be Instantiated)

Table 4 summarizes the parameters that need to be specified. We prioritize interpretability and a low number of drivers.

Table 4: Task-level model parameters and plain-language meanings.

Parameter	Typical Range/Setting	Meaning (Plain Language)
k, t_0	Scenario-specific	Adoption speed and 50% adoption time
κ_s	> 0	Speed at which substitution reduces human effort
g_i	0 (default)	Baseline functional demand growth (non-GenAI)
κ_c	0 or > 0	Strength of augmentation expanding total demand
ε	e.g., 0.05	"Negligible task" threshold (5% of baseline)
θ	e.g., 0.30	Extinction mass required to redefine profession
δ	e.g., 0.10–0.30	Task share drift threshold (JS divergence)

7.2 Visualization Plan

This section defines the charts needed to display task structure changes and redefinition evidence.

7.2.1 Figure Templates

Figure 4: GenAI adoption scenarios $A(t)$ defined by (31) (fast/medium/slow). Inflection time t_0 marks 50% adoption.

Figure 1: Adoption Scenarios $A(t)$.

Figure 5: Relative task labor demand trajectories $D_{ij}(t)/D_{ij}(0)$ for the top K tasks with highest baseline weights in profession i , across adoption scenarios. This figure directly shows which tasks become negligible and which remain core.

Figure 2: Task Labor Demand Trajectories for Top Tasks.

Figure 6: Time-varying task structure $w_{ij}(t)$ (stacked area chart or heatmap). This figure visualizes the intra-occupational reallocation driven by substitution via (27).

Figure 3: Task Structure Drift $w_{ij}(t)$.

Figure 4: Redefinition Evidence and Trigger Time t^* .

Figure 7: Redefinition evidence over time: extinction mass $M_{\text{drop}}(t)$ with threshold θ , and (optionally) $\text{JS}(w_i(t)||w_i(0))$ with threshold δ . The earliest crossing defines t^* in (36).

7.3 Sensitivity and Robustness

7.3.1 Low-Confidence Task Sensitivity

Some tasks may have low-confidence (s_{ij}, c_{ij}) assignments due to weak semantic matches with authority assessment dimensions. For each flagged task, we run two bounded sensitivity cases:

- **Deletion:** Remove the task from the set and renormalize the remaining $w_{ij}(0)$.
- **Relabeling:** Keep $w_{ij}(0)$ but replace (s_{ij}, c_{ij}) with the second-best acceptable mapping (or a conservative bound).

We report changes in: (i) whether redefinition occurs within the horizon, (ii) trigger time t^* , and (iii) the extinction mass curve $M_{\text{drop}}(t)$.

7.3.2 Key Parameter Sensitivity

We vary only a minimal, high-leverage set of parameters to avoid overfitting:

- κ_s (substitution speed): slow/medium/fast values.
- κ_c (scale-up on/off): $\kappa_c = 0$ vs. a positive value.
- Thresholds $(\varepsilon, \theta, \delta)$: conservative vs. aggressive settings.

For each combination, we summarize the stability of qualitative conclusions (which tasks become negligible, and whether/when redefinition is triggered).

7.4 Output Summary Template (Per Profession)

For each profession and each adoption scenario, we present:

1. **Task Demand Results:** List of tasks ranked by $D_{ij}(15)/D_{ij}(0)$, highlighting those below ε .
2. **Task Structure Change:** Visualization of $w_{ij}(t)$ and drift measure $\text{JS}(w_i(t)||w_i(0))$.
3. **Redefinition Decision:** Whether the criterion in (34) or (35) is met, and the trigger time t^* if applicable.

8 Simulation Results and Sensitivity Analysis

Provide visual data through charts. Discuss the robustness of the model. Sensitivity analysis is crucial for achieving Meritorious or Outstanding awards.

9 Discussion: Strengths and Limitations

9.1 Advantages

High prediction accuracy; suitable for high-dimensional data.

9.2 Shortcomings

The computational complexity of large-scale matrices is relatively high.

10 Conclusion

Summarize research findings and provide actionable recommendations.

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Appendices

Appendix A Python Implementation Code

```
1 def model_solver(data):  
2     # Core mathematical logic  
3     return results
```

AI Tools Usage Report

1. Dialogue Traceability

For each use, provide the accurate **prompt/query** and the **original AI output** (or a representative excerpt). Complete logs may be omitted only when AI is used solely for translation; in such cases, specify the **tool name and version**.

- **Example—Prompt:** "Suggest an outline for the model section." **Output (excerpt):** "1. State assumptions; 2. Define notation; 3. Derive main equations. . ."
- **Teams using translation only:** Declare the tool name and version (e.g., DeepL 4.2, ChatGPT-4); a full prompt log is not required.

2. Phased Declaration

Specify the stages at which AI is used, for example:

- **Initial Ideation/Brainstorming**
- **Code Correction/Debugging**
- **Translation/Text Editing**

Indicate whether each stage is used and specify the tools employed at each stage.

3. Risk Verification

Describe how the team checks for **hallucinations** and **incorrect citations**: for example, manual fact-checking of key claims, verifying references, cross-checking formulas and code. If used only for translation, declare that no substantive technical content was delegated to AI processing.